Final Exam Prepare - 23/9 exam

Final paper set - 202601 / 202505

Q1 - Format:Chap3 的 (8-10 mark theory & 17 mark calculate)

202601

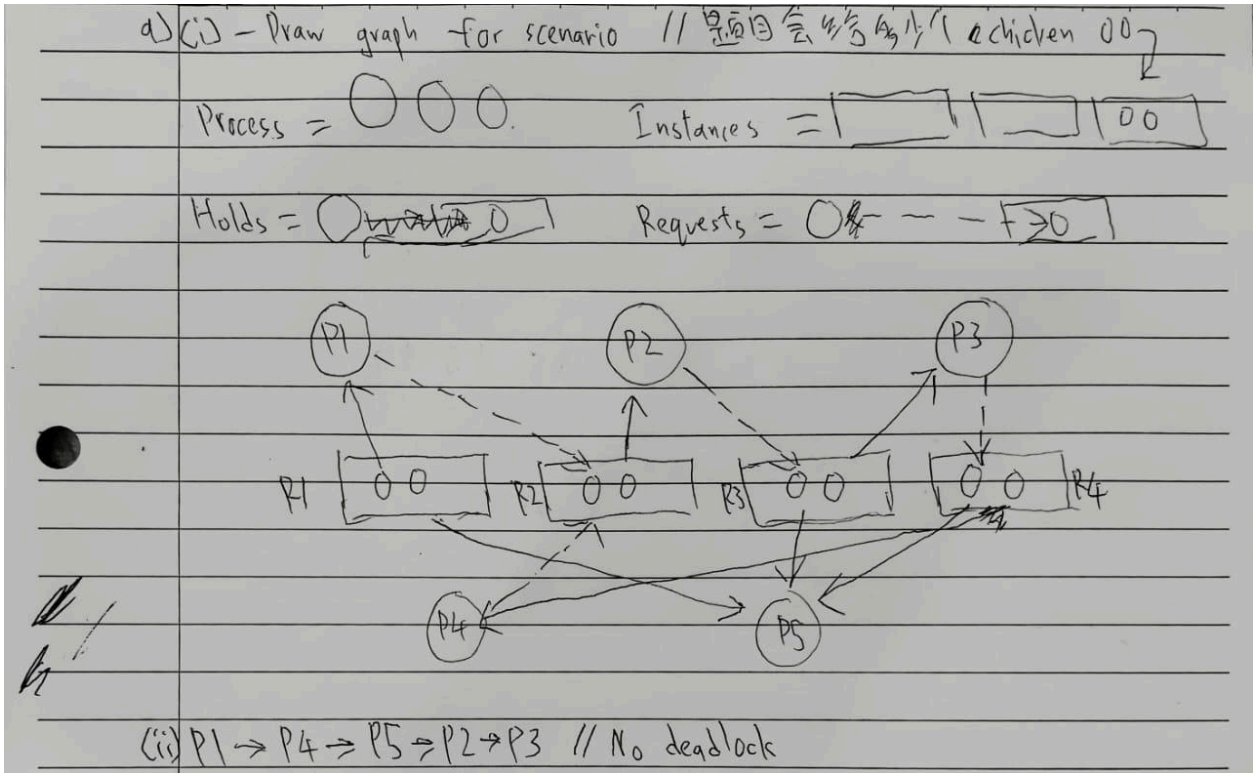
a) There are **FIVE (5)** processes (P1, P2, P3, P4 and P5) and **FOUR (4)** resources (R1, R2, R3 and R4). Assuming that all the resources are non-sharable and the number of instances for each resource is 2, 2, 2, 2 respectively.

- Process P1 holds one instance of R1. It requests one instance of R2.
- Process P2 holds all instances of R2. It requests one instance of R3.
- Process P3 holds one instance of R3. It requests one instance of R4.
- Process P4 holds one instance of R4. It requests one instance of R2.
- Process P5 holds one instance of R1, one instance of R3 and one instance of R4.

(i) Draw a directed resource allocation graph for the scenario above. (6 marks)

(ii) Is the system in a deadlock state? Justify your answer. (3 marks)

a) (i) 这边的ans错了



(ii) No deadlock, P1-> P4 -> P5 -> P2 -> P3

b) Assuming that there are **FOUR (4)** processes (P1, P2, P3 and P4) being processed by a CPU. The processes need **THREE (3)** resources (A, B and C). The information shows the current resources allocated to each process and also the maximum resources they require in order to complete their processes in Table 1. Given that the available resources are 2, 2, 2 respectively.

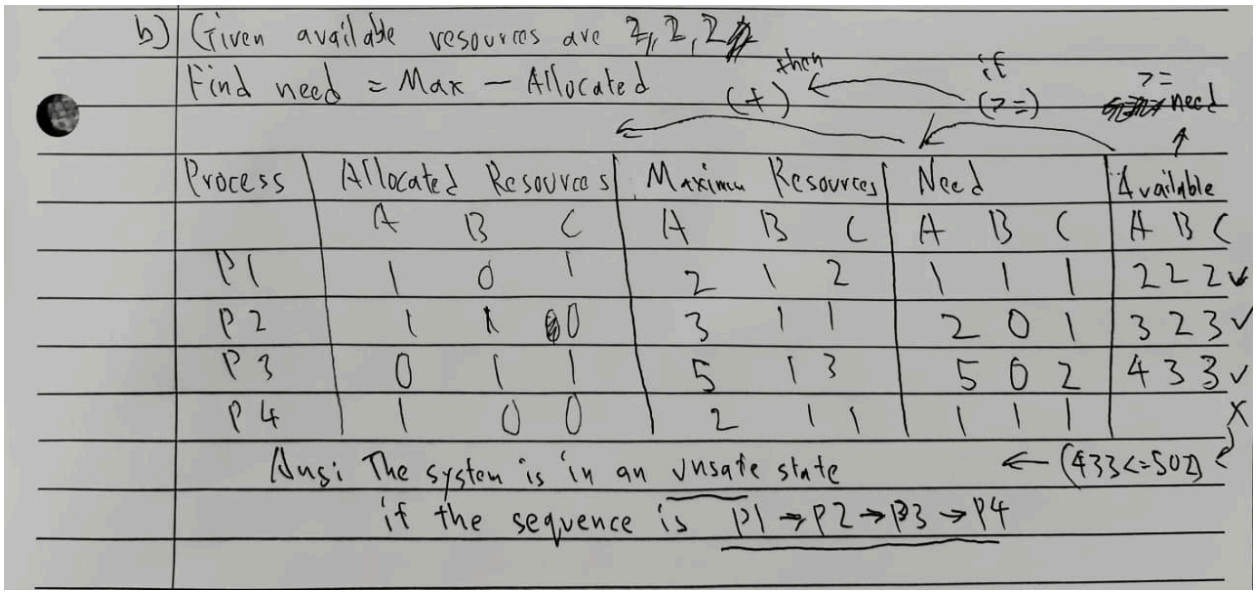
Processes	Allocated Resources			Maximum Resources Required		
	A	B	C	A	B	C
P1	1	0	1	2	1	2
P2	1	1	0	3	1	1
P3	0	1	1	5	1	3
P4	1	0	0	2	1	1

Table 1

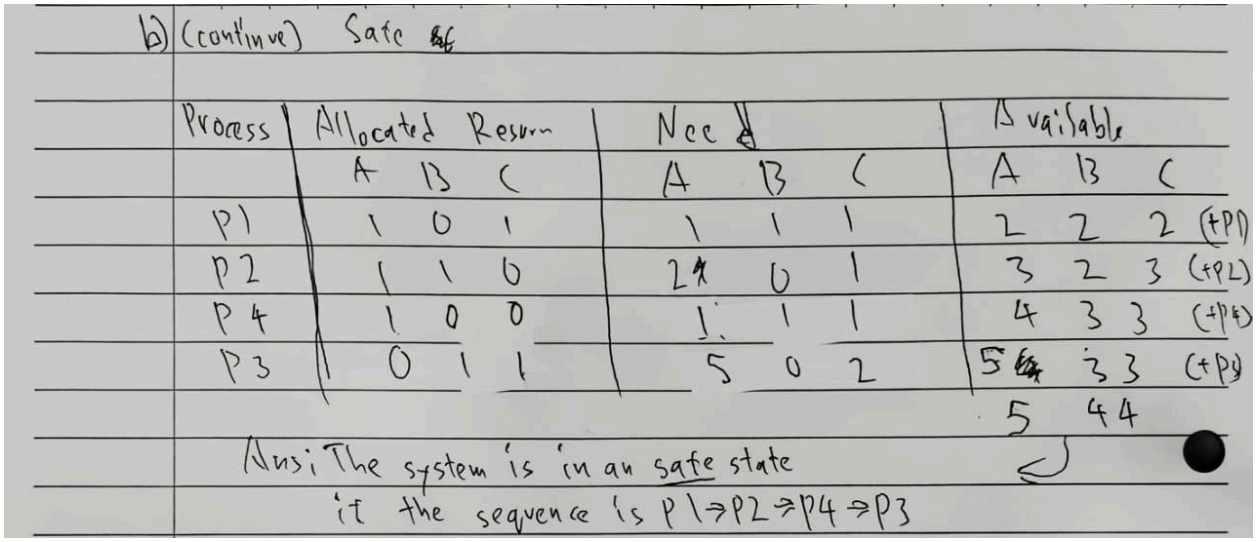
Use ***Banker's Algorithm*** to determine whether the system is in a safe state or unsafe state when the process sequence is P1 to P4. Demonstrate a scenario in which the system is in an unsafe state, and propose a process sequence that would restore it to a safe state. Justify your answer.

(8 marks)

b)



Explain format: The system is in a/an unsafe/safe state if the sequence is P1->???



b) (continue) Safe

Process	Allocated Resources			Need			Available		
	A	B	C	A	B	C	A	B	C
P1	1	0	1	1	1	1	2	2	2 (+P1)
P2	1	1	0	2	0	1	3	2	3 (+P2)
P4	1	0	0	1	1	1	4	3	3 (+P4)
P3	0	1	1	5	0	2	5	3	3 (+P3)
							5	4	4

Ans: The system is in an safe state
if the sequence is P1 ⇒ P2 ⇒ P4 ⇒ P3

202505

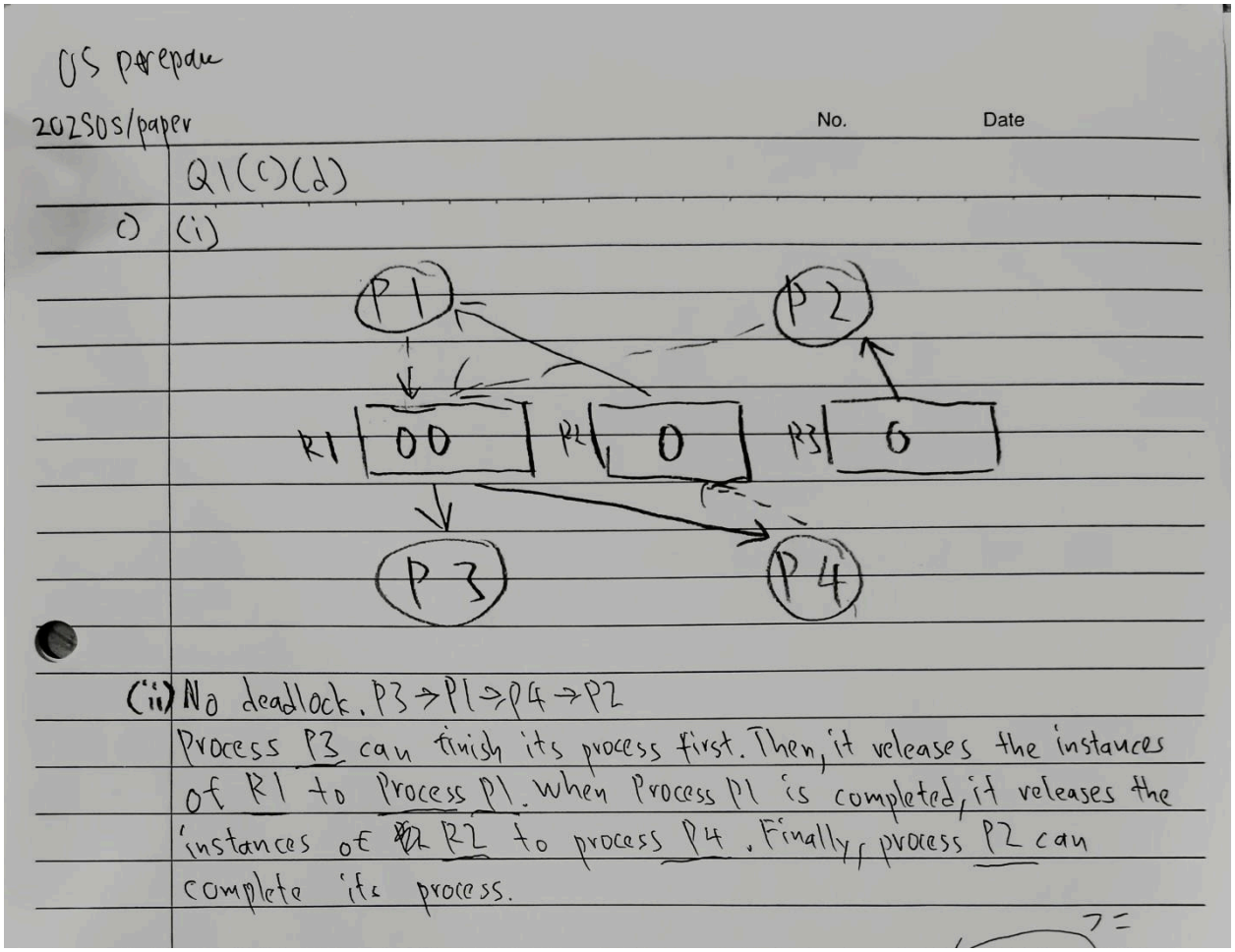
c) Consider a system of **FOUR (4)** processes (P1, P2, P3, P4), **THREE (3)** non-sharable resources (R1, R2, R3) and the number of instances for each resource are 2, 1, 1 respectively.

- P1 holds an instance of R2 and requests an instance of R1.
- P2 holds an instance of R3 and requests an instance of R1.
- P3 holds an instance of R1.
- P4 holds an instance of R1 and requests an instance of R2.

(i) Draw a directed resource allocation graph based on the above scenario. (5 marks)

(ii) Is the system in a deadlock state? Justify your answer. (3 marks)

c) (i)



(ii) Format:
Process (1) can finish its process first. Then, it releases the instances of (??) to Process (2). When Process (2) is completed, it releases the instances of (??) to Process (3). (...). Finally, Process (4) can complete its process.

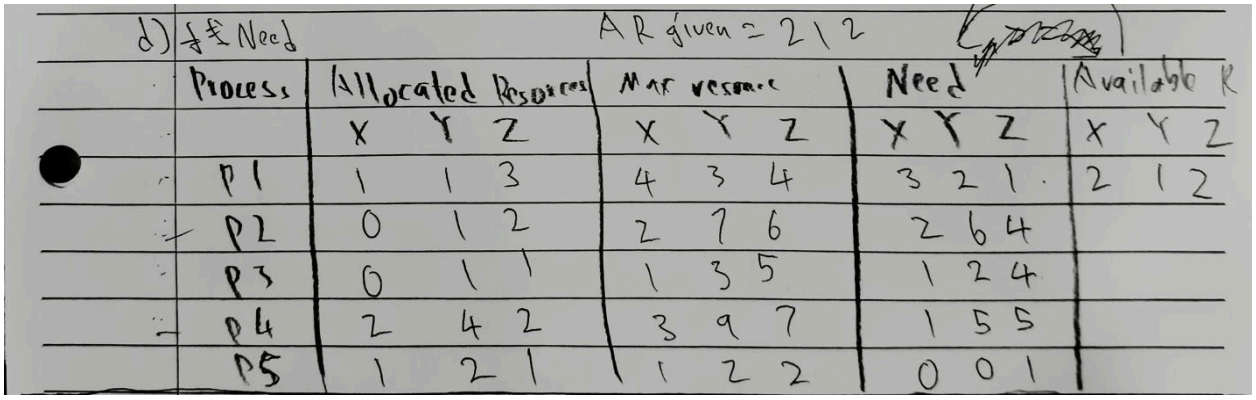
d) Assuming there are **FIVE (5)** processes (P1, P2, P3, P4, and P5) which need **THREE (3)** resources (X, Y, and Z). **Table 1** shows the current resources allocated to each process and the maximum resources required by each process. Given that the available resources are 2, 1, 2 respectively.

Table 1

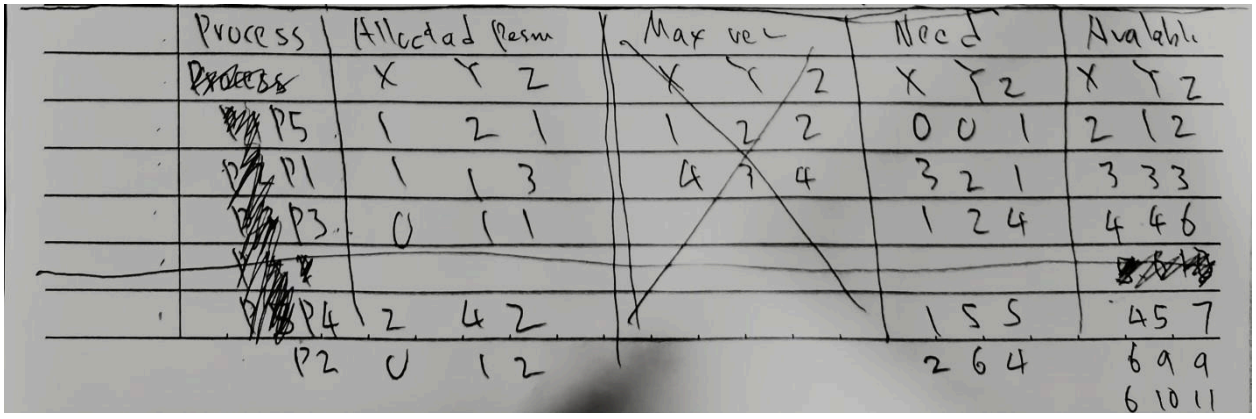
	Allocated resources			*Max resources required*		
	X	Y	Z	X	Y	Z
P1	1	1	3	4	3	4
P2	0	1	2	2	7	6
P3	0	1	1	1	3	5
P4	2	4	2	3	9	7
P5	1	2	1	1	2	2

By using the *Dijkstra's Banker's algorithm*, determine whether the system is in a safe or unsafe state. Justify your answer by showing the progress of resources availability and determine the *Need* Matrix of each of the process. (7 marks)

d)



The system is in an unsafe state if the sequence is P1->P2->P3->P4->P5



The system is in a safe state if the sequence is P5->P1->P3->P4->P2

Theory part

Q(c) In operating systems, a deadlock may cause processes from making progress and might require external intervention to resolve. Explain why a deadlock cannot occur if processes hold only one type of resource. (4 marks)
Ans:
Deadlock requires a circular wait, where each process holds one resource and waits for another resource held by another process.
If a process can hold only one type of resource, it cannot hold one resource while waiting for another.
Therefore, hold and wait cannot occur, and no circular wait can be formed. Since one of the four necessary conditions is missing, deadlock cannot occur.

Q(d) In operating systems, deadlock is generally considered more severe than starvation. Explain the concept of starvation and describe how it differs from deadlock. (4 marks)
Ans at 4. Deadlock VS Starvation

1. Explain the meaning of deadlock. (Page 2)
Deadlock is when two or more processes are blocked forever because each holds a resource and waits for another held by another process

2. Discuss the FOUR (4) conditions for deadlock. (Page 13-14)
Mutual Exclusion
- Resource is in non-sharable mode

Hold and Wait
- A process holds one resource while waiting for another resource.

No Preemption
- Resources cannot be forcibly taken away.

Circular Wait
- Process A waits for Process B, B waits for C, and the last process waits for A, forming a circular chain.

3. Strategies for Handling Deadlocks - Four strategies (Page 24)
Prevention
- Prevent occurrence of one condition.
Avoidance
- Avoid deadlock if it becomes probable.
Detection
- Detects deadlock when it occurs.
Recovery
- Resume system normalcy quickly and gracefully.

4. Deadlock VS Starvation

	Deadlock	Starvation
Concept	Deadlock is when two or more processes are blocked forever because each holds a resource and waits for another held by another process	Starvation is when one process waits indefinitely for a resource because it is never chosen to get it
Different		
1. Affected processes	Two or more processes are blocked forever.	One or a few processes cannot get resources.

Q2 - Format:Chap 4 的 (0-2 mark theory & 23-25 mark calculate)

202601

a) Consider the following page reference string:

2	9	3	6	7	2	0	0	6	3	0	1

Assuming that **FOUR (4)** memory frames are initially empty in the memory, perform the page faults trace analysis to determine the *number of page faults* that will occur for each of the following page replacement algorithms.

(i) First-In-First-Out (FIFO) (4 marks)

(ii) Least-Recently-Used (LRU) (4 marks)

(iii) Optimal Page Replacement (4 marks)

a)

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Q2

a) 2 9 3 6 7 2 0 0 6 3 0 1 最小需要多少?

4個位 ① ② ③ ④

(i) FIFO: (最高FIFO) Format: copy過去

Format ↗

2 9 3 6 7 2 0 0 6 3 0 1

①	2	2	2	2	7	7	7	7	7	7	1
②		9	9	9	9	2	2	2	2	2	2
③			3	3	3	3	0	0	0	0	0
				6	6	6	6	6	3	3	3
	F	F	F	F	F	F		F		F	

有空的就放F

No. of Page fault = 9 → 有有多少個F

Hit ratio = 9/12 = 0.75

(ii) LRU: (看回去,拿排的) (拿最近的) 換

2 9 3 6 7 2 0 0 6 3 0 1 ☆

	2	2	2	2	7	7	7	7	3	3	3
		9	9	9	9	2	2	2	2	2	1
			3	3	3	3	0	0	0	0	0
				6	6	6	6	6	6	6	6
	F	F	F	F	F	F	F		F		F

No. of Page fault = 9

(iii) Optimal: (看前面的, 拿最遠的) 換

2 9 3 6 7 2 0 0 6 3 0 1

	2	2	2	2	2	2	0	0	0	0	0	1
		9	9	9	7	7	7	7	7	7	7	7
			3	3	3	3	3	3	3	3	3	3
				6	6	6	6	6	6	6	6	6
	F	F	F	F	F		F					F

如果後面有2個都不會在,拿什麼都可以

No. of Page fault = 7

b) Given the following information:

Free Space List	
Partition	Size (KB)
A	150
B	280
C	200
D	350
E	250

Queuing Processes	
Process	Size (KB)
P1	200
P2	90
P3	300
P4	125
P5	220

Show how the **FIVE (5)** processes are allocated using *fixed memory partitions techniques* when the following allocation algorithms are applied. Use the following template to show the working of your answers.

Table format:

Partition	Partition Size	Process	Process Size	Internal Fragmentation

(i) First-fit (4 marks)

(ii) Best-fit (4 marks)

b)

No. Date

Q2 每用一个partition，就分给它

b) Formati 从头开始找，第一个放得下的 partition 就进去 / 一个process 有多table给的 一有process 是的 结束

(i) First-fit

第一个适的

Partition	Partition Size	Process	Process Size	Internal Fragmentation
A	150	P2	90	60
B	280	P1	200	80
C	200	P4	125	75
D	350	P3	300	50
E	250	P5	220	30

一定要A→B→C→D→E // 选择是能放的Process先用给前面的 Partition，如果后面没有了就空着

看Process的时候也是从头start // Total Internal Fragmentation = 60+80+75+50+30 = 295

(ii) Best-fit

最适合的

Partition	Partition Size	Process	Process Size	Internal Fragmentation
A	150	P2	90	60
B	280	P5	220	60
C	200	P1	200	0
D	350	P3	300	50
E	250	P4	125	125

Total Internal Fragmentation = → C

找时用个子能放的最小 partition // choose 最小 但 enough 的 (Process)

(iii) Worst-fit (最差的)

Partition	Partition Size	Process	Process Size	Internal Fragmentation
A	150			
B	280	P2	90	190
C	200			
D	350	P1	200	150
E	250	P4	125	125

找最大的 Partition，if 和process match // P3和P5会空

Exp: P1=200, B,C,D,E 有B match // 看最大的 → D(350)

c) Given a system with a 64 GB logical address space and a page size of 8 MB, where logical addresses are mapped on to a 128 GB physical memory. Determine the format of the logical address and physical address, showing every component and the size of each component.

(5 marks)

c)

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Q2

c) Given a system ... 64GB logical address space ... a page size of 8 MB, ... to a 128GB physical memory. Determine ... format of the logical address and physical addres

Logical Address = Page number + Page Offset
physical Address = Frame Number + Frame Offset

Step1 Find Logical address have many bits // 1 GB = 2^{30} bytes
Format = $2^{?}$ = ___ (64GB)
64GB = 64×2^{30}
$= 2^6 \times 2^{30} = 2^{36}$ = Logical addres = 36 bits

Step2 Find Page Offset // 1 MB = 2^{20} bytes

8 MB = 8×2^{20}
$= 2^3 \times 2^{20} = 2^{23}$ bytes = 23 bits

Step3 Page number = 36 (Logical address) − 23 (page offset) = 13 bits

Logical Address	Page number	Page Offset
36 bits	13 bits	23 bits

Step4 Find Physical Address bits // 1 GB = 2^{30} bytes
128GB × 2^{30}
= $2^7 \times 2^{30} = 2^{37}$ bytes = 37 bits

Step b − Frame Offset
Per page size - 幀
= 8 MB
(page size) = 2^{23} byte
= 23 bits

Step5 Find Number of Frames = Physical Memory / Frame Size
= 128GB / 8MB
= 128GB × 1024MB = 131072 MB
= 131072 MB / 8 MB = 16384
= 2^{14}
= 14 bits

Pyssical address	Frame Number	Frave Offset
37 bit	14 bits	23 bits

202505

a) Consider the following page reference string:

1	2	3	4	2	5	2	6	4	5	2	5	1

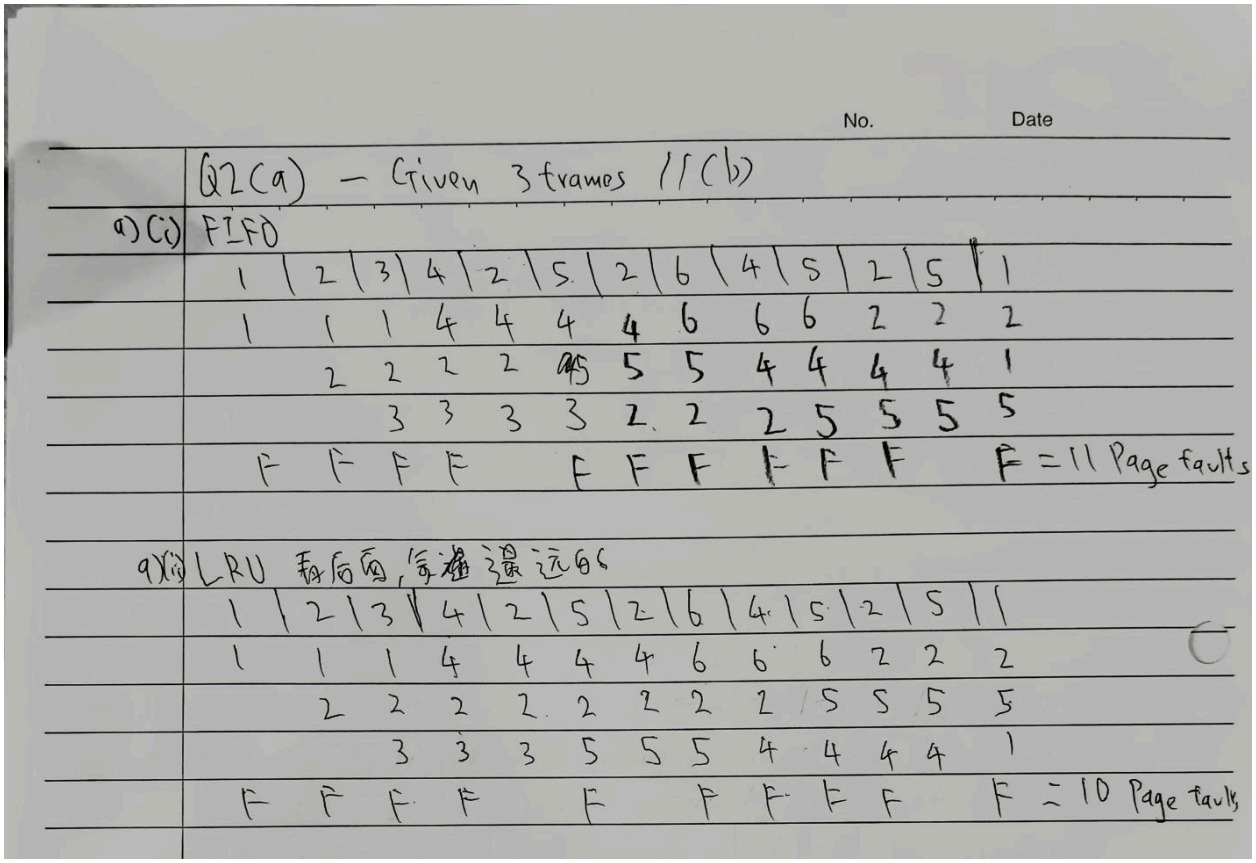
Assuming that **THREE (3)** frames are initially empty in the memory. Perform a page trace analysis and calculate the number of page faults using each of the following page replacement algorithms:

(i) First-In-First-Out (FIFO) (4 marks)

(ii) Least-Recently-Used (LRU) (4 marks)

a) (i)

a) (ii)



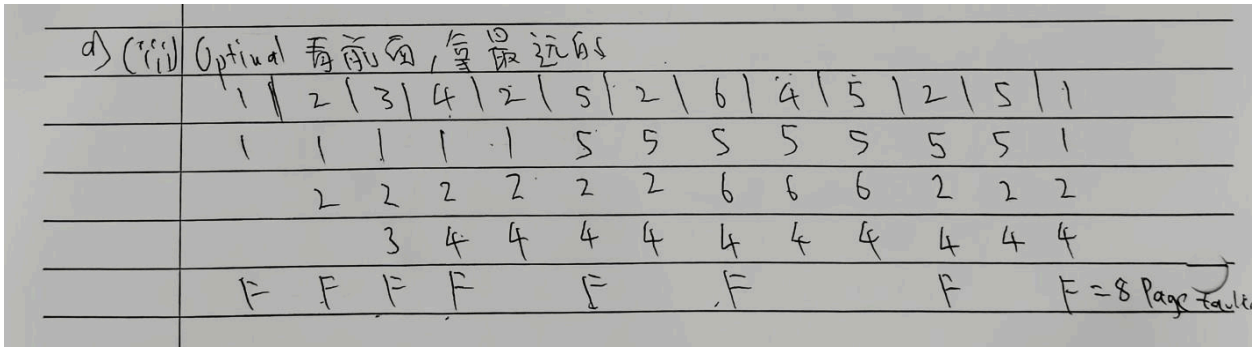
Question 2 a) (Continued)

(iii) Optimal Page Replacement (4 marks)

(iv) Based on your answers for **Question 2 a) (i)** to **Question 2 a) (iii)**, identify the page replacement algorithm that demonstrates the best performance. (1 mark)

a) (iii)

a) (iv)



b) Given the following information in **Table 2**.

Table 2

Free Space List	
Partition	Size (KB)
A	35
B	45
C	40
D	60

Queuing Processes	
Process	Size (KB)
P1	51
P2	39
P3	43
P4	29

Refer to **Table 2**, show how the **FOUR (4)** processes are allocated into memory partitions using the following memory allocation algorithms.
Note: Use the following table format to show your answers. If a wait situation occurs, indicate which process need to wait.

(i) First-Fit (4 marks)

(ii) Best-Fit (4 marks)

b)

(b) (i) First-Fit Fix memory

Partition	Partition Size	Process	Process Size	F
A	35	P4	29	6
B	45	P2	39	6
C	40	—	—	—
D	60	P1	51	9

P3 process need to wait.

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Q2(b) (cont.)

(ii) Best-Fit // 拿最靠近的

Partition	Partition Size	Process	Process Size
A	35	P4	29
B	45	P3	43
C	40	P2	39
D	60	P1	51

c) Consider a system with a logical address space of 64 pages of 2K each, mapped onto a physical memory of 512 frames. Determine how many bits are there in the *logical address* and *physical address*.
(Hint: You are required to show the calculation steps in your answer.) (4 marks)

c)

c) 題目 Given logical address = 64 page & 2K

physical address = 512 & 2K

Logical address = Page number + page offset

6 + 11 = 17 bits

① Find page number

$64 = 2^6 \Rightarrow$ 6 bits // Page number = 6

② Find page offset

$2K = 2 \times 1024 = 2048 = 2^{11} \Rightarrow$ 11 bits

Offset = 11

Physical address = Frame number + Frame offset

9 + 11 = 20

① Find Frame number

$512 = 2^9 \rightarrow$ 9 bits // Frame number = 9

② Find Frame offset

$2K = 2 \times 1024 = 2048 = 2^{11} = 11$ bits

Offset = 11

Ans:

Logical address = 17 bits

Physical address = 20 bits

Q3 - Format:Chap 5 的 (15-17 mark 的 theory & 8-10 mark 的 calculate)

202601

a) Rewrite the following arithmetic expression to take advantage of concurrent processing by using the terms **COBEGIN** and **COEND** to delimit the sections of concurrent code.

$$V = (A + B * C) * (C + D - E / F)$$

(8 marks)

a)

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Q3

a) Rewrite the ---- -- by using COBEGIN and COEND ---- (8 marks)

V=(A+B*C)*((+D-E/F)

Cobegin
T1=B*C
T2=E/F
Coend

這兩行平行執行, No dependency

Cobegin
T3=A+T1
T4=D-T2
COEND

T3=(A+T1(B*C))
T4=(D-T2(E/F))

T5=C+T4
V=T3*T5

Extra(b) A couting semaphore S is initially set to ?. Determine the final value of S after performing ? P operation followed by ? V operations. Explain how you obtain the answer.

Formati: P operation = decrease 1 / V = increase 1 / Final S = S-(P)+(V)

Ans: P(S) → S = S-1 = 6-1 | V(S) → S = 5+1 → 4+1 = 5

Given S=6 P=2 =5-1=4 =6+1 S=5

P=2
V=1

Final value of S = 5

Expl.: Initially, the semaphore value is S=6. When performing the first P operation, the value of S is decreased by two. Therefore, S=4

Next, one V operations are performed. Each V operation increases the semaphore value by 1 ove. Thus, the value of S = 4+1 = 5.

The Final value of S is 5

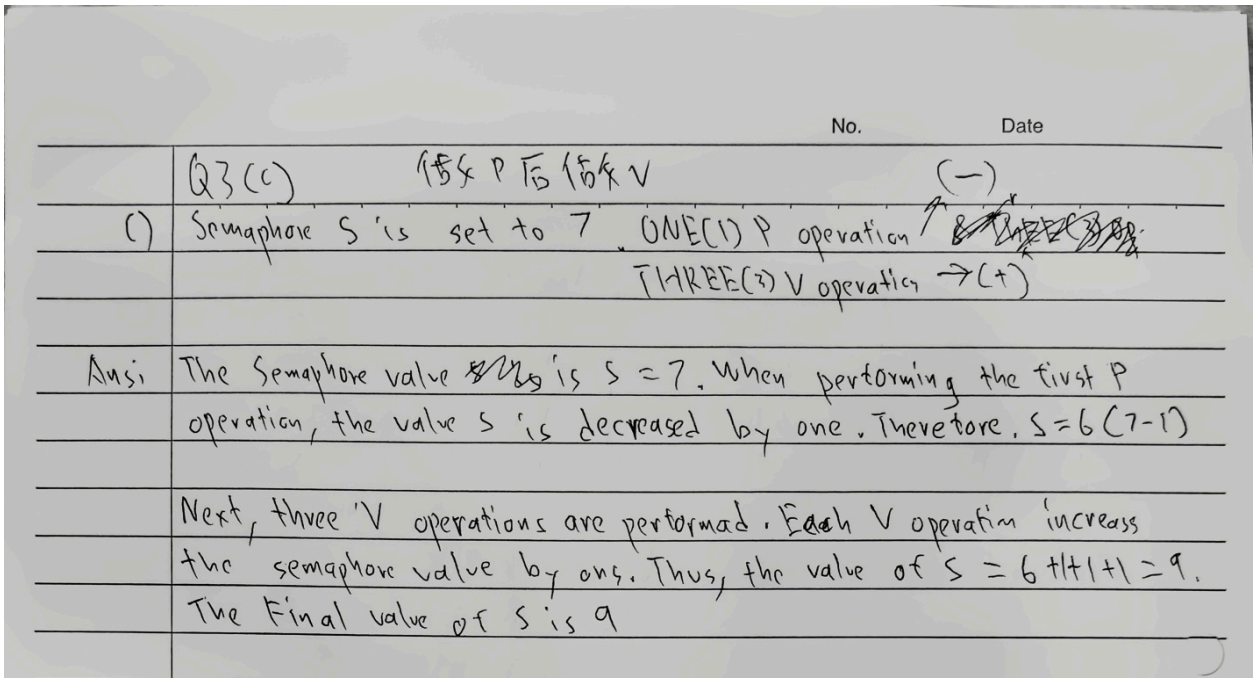
202505

c) A counting semaphore S is initially set to 7. Determine the final value of S after performing **ONE (1)** ***P*** operation followed by **THREE (3)** ***V*** operations. Explain how you obtain the answer. (5 marks)

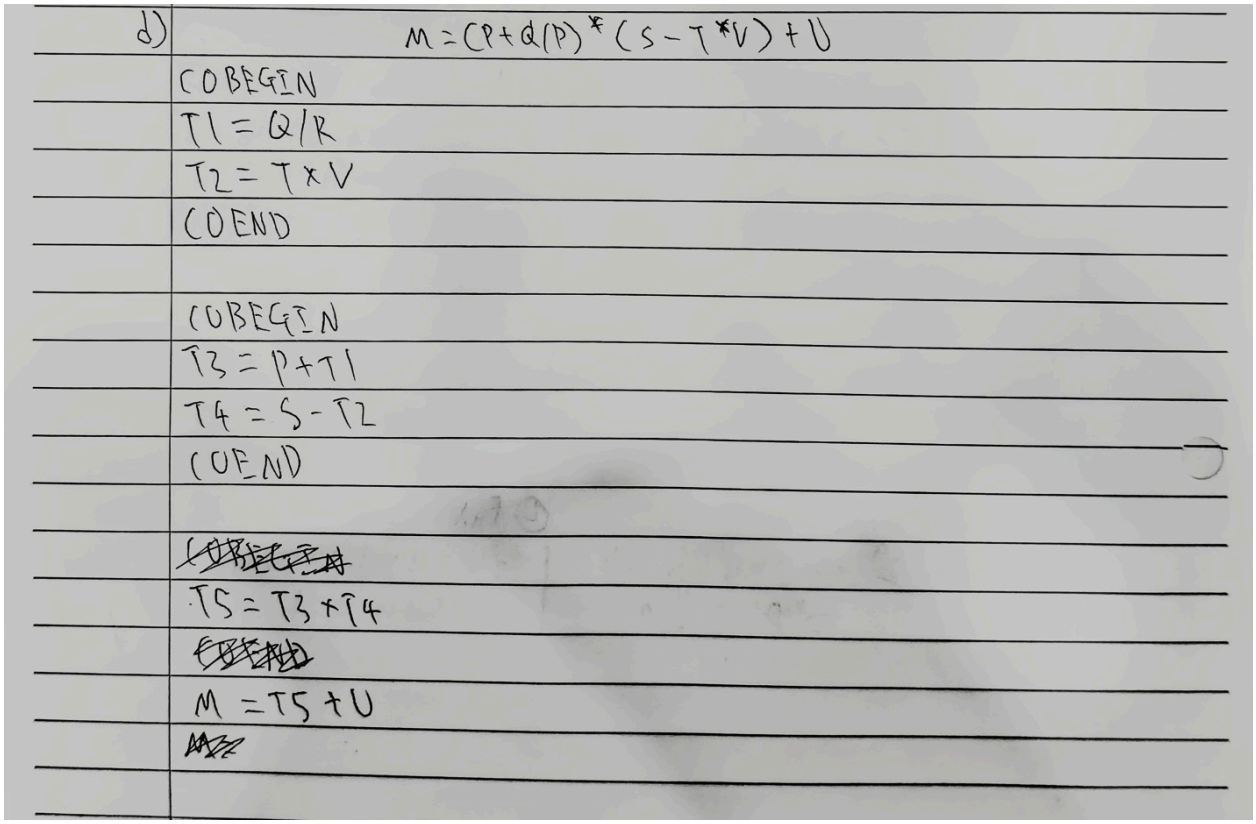
d) Rewrite the following arithmetic expression into concurrent code that utilises concurrent processing. Use **COBEGIN** and **COEND** to define the concurrent sections.

$$\mathbf{M = (P + Q / R) * (S - T * V) + U}$$ (5 marks)

c)



d)



Theory part

1. Parallel Processing (Page 2–3)

Definition

- Parallel processing (multiprocessing) means 2 or more processors operate in parallel.

Two Benefits

- Increased reliability – if one processor fails, others take over.
- Faster processing – instructions can be processed in parallel, two or more at a time.

Two Challenges

- How to connect the processors into configurations.
- How to orchestrate their interaction.

2. Critical Region (Page 5–6)

Definition

- A critical region is a section of a program that must complete its execution before other processes can access the shared resources it uses.

Importance

- Ensures mutual exclusion, one process cannot access the critical region while another is inside.

Lock-and-Key Mechanism

- Synchronization is implemented as a lock-and-key arrangement.

3. Semaphore (Page 7–9)

Definition

- A semaphore is an integer value used to provide mutual exclusion.
- Many problems can be solved using binary semaphores, which take values 0 or 1.

Two Operations

- wait(s) — also called P(s) or down(s)
- signal(s) — also called V(s) or up(s)

How wait and signal work

wait(S):

If $S \leq 0$, the process blocks (waits).

Otherwise, S is decremented by 1 and the process proceeds.

signal(S):

Unconditionally increments S by 1.

Wakes up a waiting process if any.

4. Readers and Writers（Page 14–17)

- Readers: only read existing data; many readers can be active at the same time.
- Writers: modify existing data; must be carefully accommodated.

Q4 - Format: Chap6&7 的 (9-17 mark theory) & Chap 6 的 (8-16 mark calculate)

202601

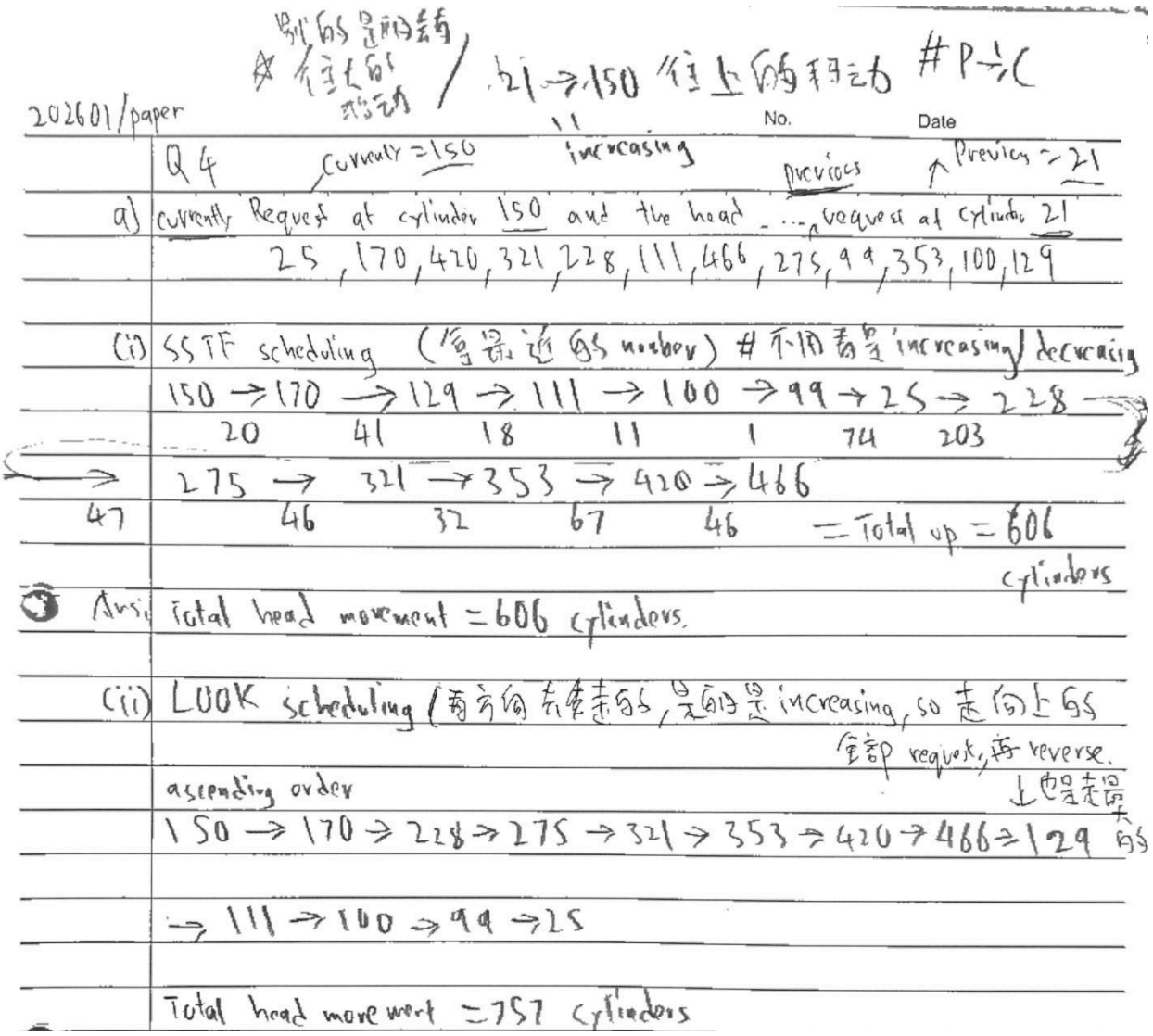
a) A disk drive has 500 cylinders, numbered from 0 - 499. The drive is currently serving a request at cylinder 150 and the head was previously serving a request at cylinder 21. The queue of pending requests is in the following order:

25, 170, 420, 321, 228, 111, 466, 275, 99, 353, 100, 129

By referring to the above queue of pending requests, calculate the total head movements using the following disk seek scheduling:

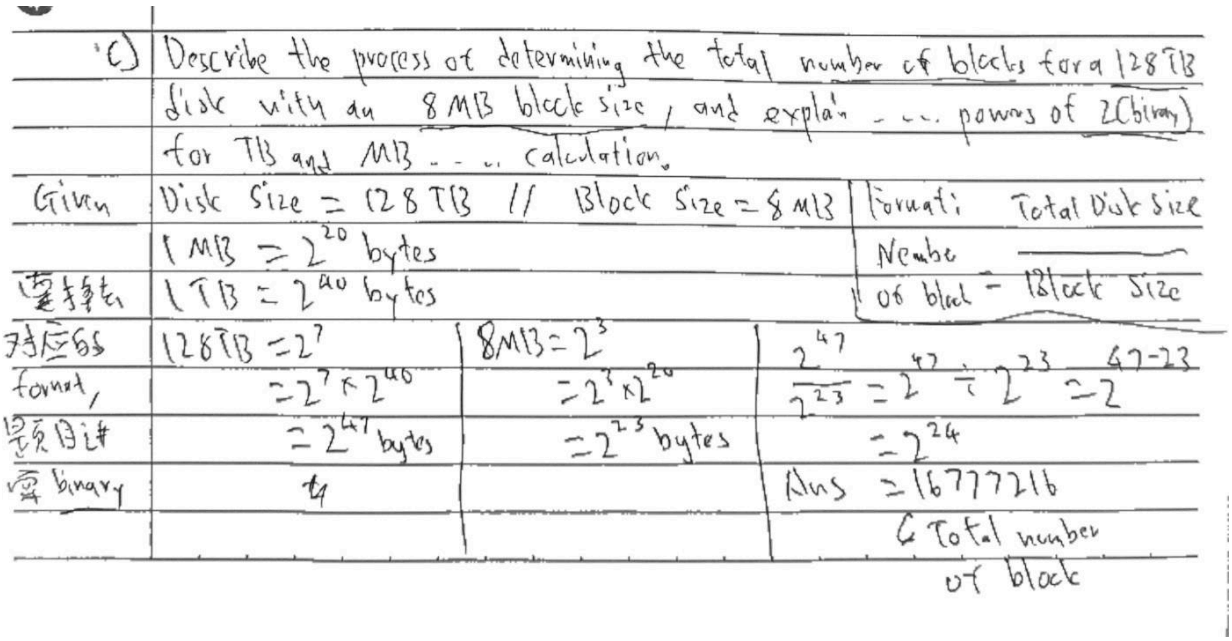
(i) SSTF scheduling (4 marks)

(ii) LOOK scheduling (4 marks)



c) Describe the process of determining the total number of blocks for a 128 TB disk with an 8 MB block size, and briefly explain how standard powers of 2 (binary) for TB and MB are applied in the calculation. (5 marks)

c)



202505

a) A SATA hard disk contains 200 cylinders numbered from 0 to 199. The disk head is currently at cylinder 90, and it previously served a request at cylinder 60. The pending I/O request queue is:

40, 10, 133, 170, 19, 75, 180, 150

Calculate the total head movements using the following disk scheduling algorithms.
Note: Show your working clearly using head movement sequences.

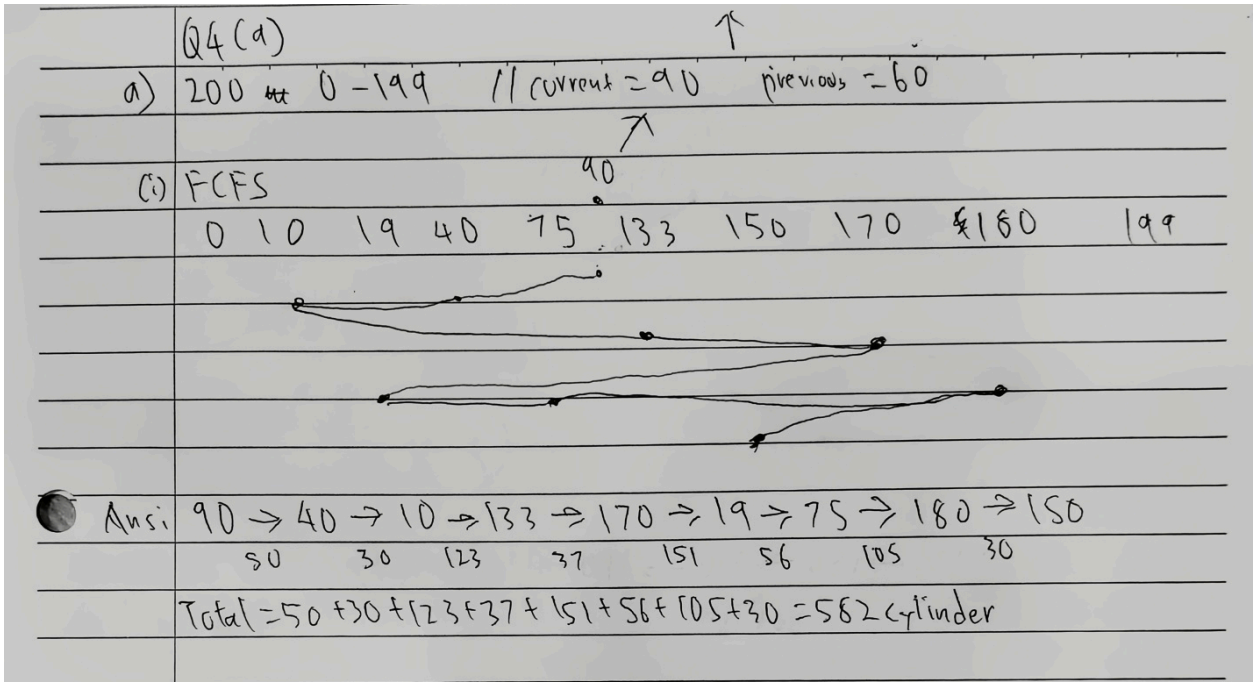
(i) FCFS scheduling (4 marks)

(ii) SSTF scheduling (4 marks)

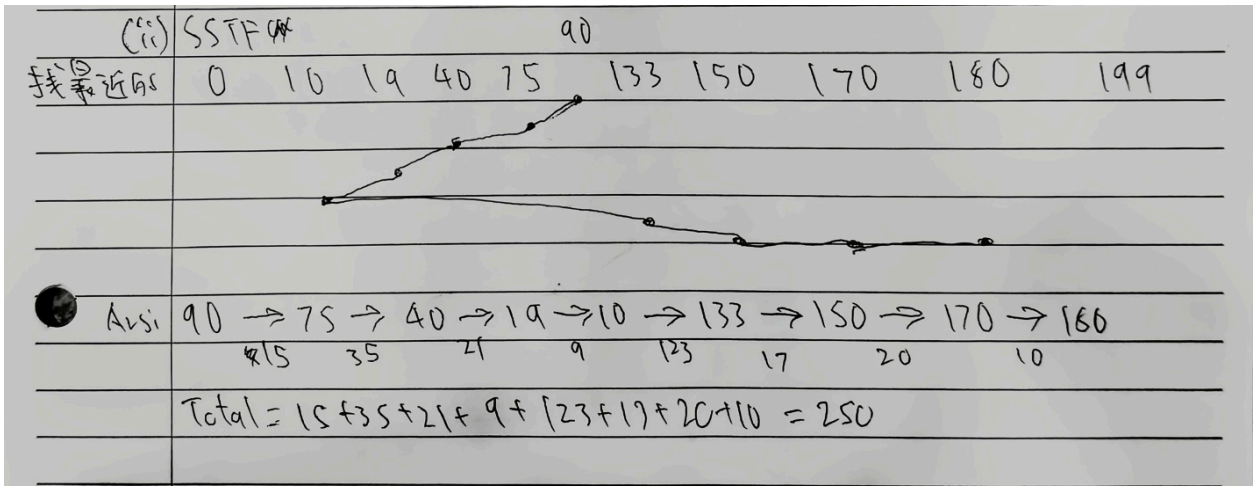
(iii) SCAN scheduling (4 marks)

(iv) LOOK scheduling (4 marks)

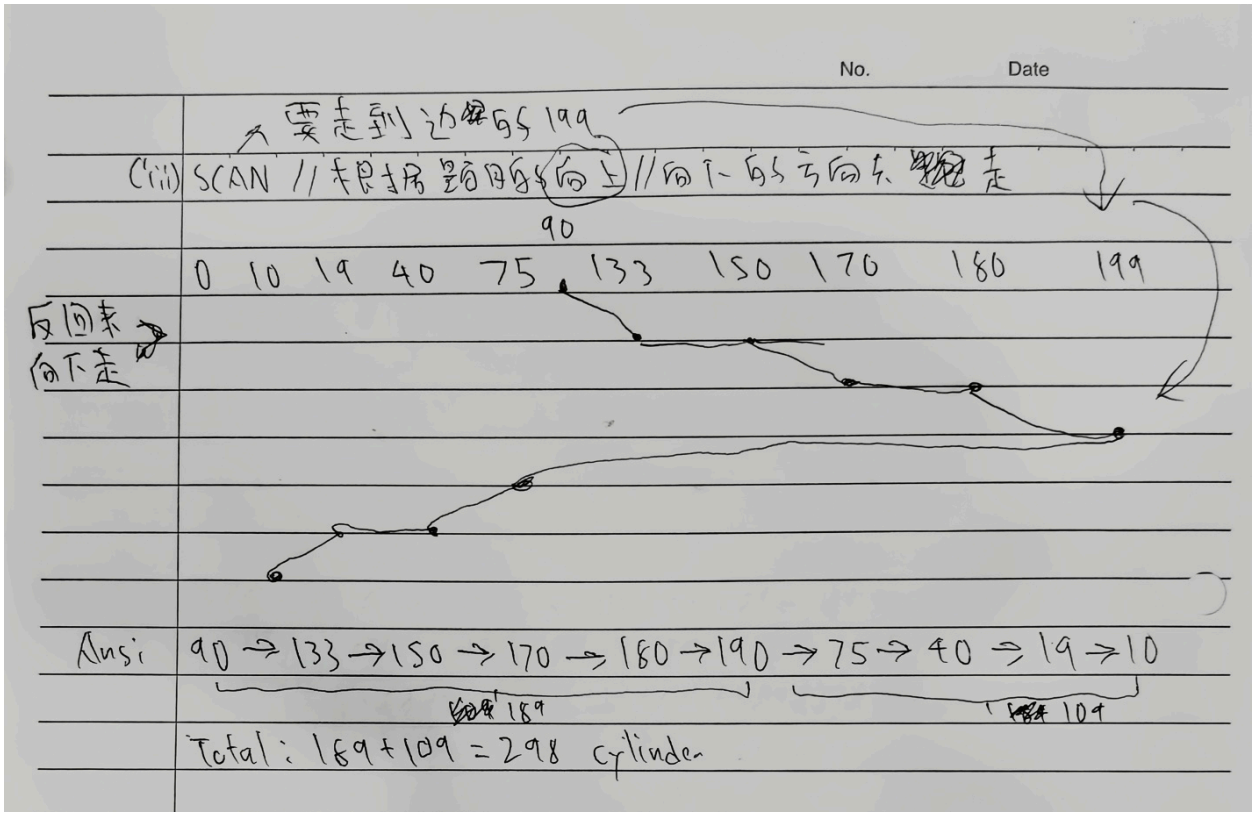
a) (i)


Q4(a)
a) 200 0-199 // current = 90 previous = 60
90
(i) FCFS
0 10 19 40 75 133 150 170 180 199
Ans: 90 → 40 → 10 → 133 → 170 → 19 → 75 → 180 → 150
50 30 123 37 151 56 105 30
Total = 50 + 30 + 123 + 37 + 151 + 56 + 105 + 30 = 582 cylinder

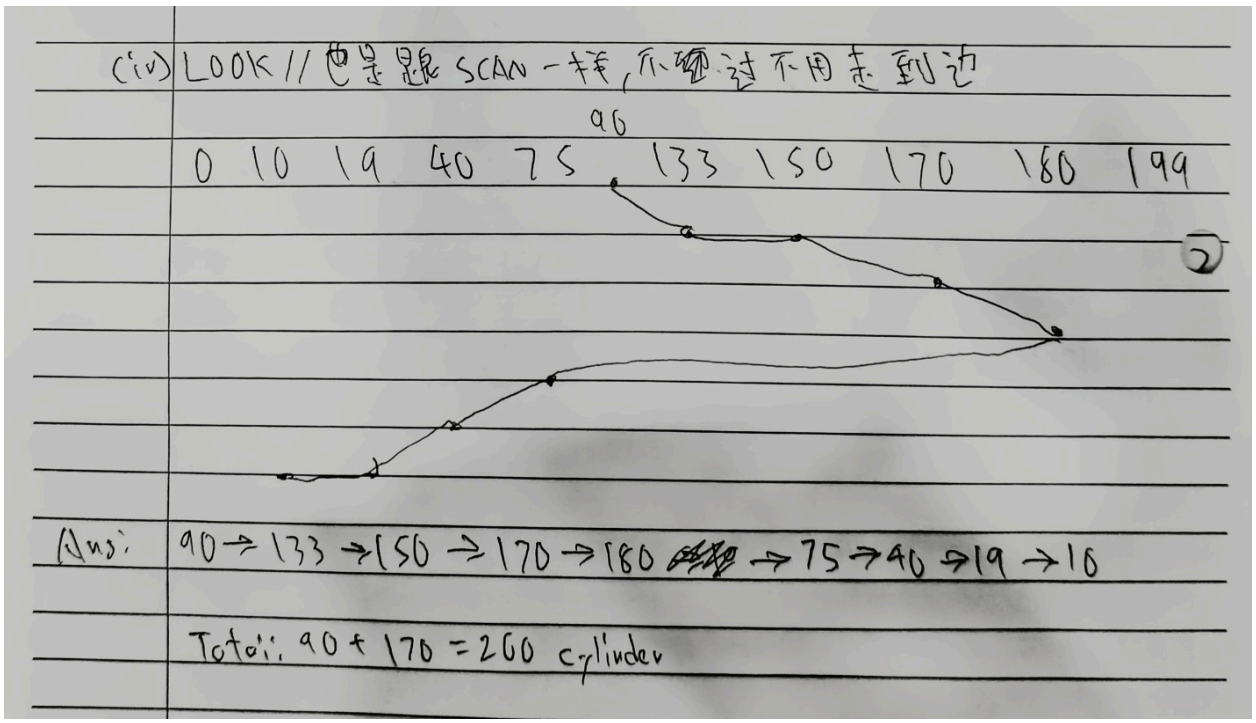
a) (ii)


(ii) SSTF
90
我最近的
0 10 19 40 75 133 150 170 180 199
Ans: 90 → 75 → 40 → 19 → 10 → 133 → 150 → 170 → 180
15 35 21 9 123 17 20 10
Total = 15 + 35 + 21 + 9 + 123 + 17 + 20 + 10 = 250

a) (iii)


No.
Date
要走到边缘的199
(iii) SCAN // 根据题目的向上 // 向下的方向走
90
0 10 19 40 75 133 150 170 180 199
反回来向下走
Ans: 90 → 133 → 150 → 170 → 180 → 190 → 75 → 40 → 19 → 10
189
104
Total: 189 + 109 = 298 cylinder

a) (iv)


(iv) LOOK // 也是跟SCAN一样，不过不用走到边
90
0 10 19 40 75 133 150 170 180 199
Ans: 90 → 133 → 150 → 170 → 180 → 75 → 40 → 19 → 10
Total: 90 + 170 = 260 cylinder

Theory part (Chap7 only)

1. File Manager 的 Responsibilities (Page 2–3)

考试常问：
“What are the responsibilities of the File Manager?”

- Track where each file is stored.
- Determine where and how files will be stored.

2. File Structure (Page 4)

Field → Record → File → Database

3. File Organization - three type(Page 9–19)

Sequential Record Organization VS Direct File Organization VS Indexed Sequential Record Organization (Page 10 - 19)

	Sequential Record Organization	Direct File Organization	Indexed Sequential Record Organization
records stored	Records stored & retrieved serially, one after another.	Uses direct access files, only on direct access storage devices.	Records stored sequentially
Advantages	Easy to implement.	Fast record access.	No collision

4. Physical Storage Allocation - three type (Page 20–29)

考试常问：
Describe THREE (3) file allocation methods.
Differentiate between contiguous, linked, and indexed allocation.

(1) Contiguous Allocation
Records stored one after another.

(2) Linked Allocation
Files use any available disk storage space.

(3) Indexed Allocation
Allows direct record access.

5. Free-Space Management- four type (Page 30–37)

考试常问：
Describe FOUR (4) free-space management methods.

Bit Vector — each block represented by 1 bit (1 = free, 0 = allocated).

Linked List — free blocks linked together.

Grouping — store addresses of n free blocks in the first free block.

Counting